

MINASA · みなさ

Socially Intelligent Support Infrastructure for Neurodivergent Children

みんなでささえる発達特性

Summary

From Concerns to Connections for More Children

What if **connections** are the key to helping neurodivergent children thrive?

No single service or specialist can meet their complex, dynamic needs alone.

8.8% of children in Japan show neurodevelopmental challenges (e.g., ADHD, LD, ASD) in classrooms — yet 43% do not receive accommodations. 2.6 years can go by from first concern to specialist care. Unmet needs often cascade into depression, school refusal, and even suicide.

A major roadblock is the shortage of specialists, and Japan's aging society won't grow them. The path forward is to **activate and coordinate available resources** — community programs, peer networks, and digital tools. However, today, no one can see them all, connect to them easily, or judge the fit well.

MINASA builds a **cyber-physical-social** ecosystem that enables socially intelligent, specialist-informed, adaptive coordination. It provides tailored, emotionally tuned, culturally and locally grounded introductions and warm handoffs that go beyond a list of resources. When parents, teachers, and frontline workers feel "something's off," MINASA turns worries and questions into a clear picture of need and the next step.

MINASA **connects each child to the right mix of supports in days** instead of years and continues to guide them as their needs change. It load-balances specialist and non-specialist resources and increases the community's capacity to support neurodivergent children — unlocking their potential and productivity.

Goal: Care implemented in under 12 weeks and 3x more children supported by year 5.

The Challenge

THE PROBLEM

Potential & productivity are lost.

- Teachers identify 1 in 11 children with neurodevelopmental concerns.
- Absence of, or long delays before, support implementation.
- Academic, social, and mental-health problems compounded, resulting in withdrawal and non-participation.
- Family and societal burdens increase.

WHY NOW

Technology is ripe. Policy is aligned.

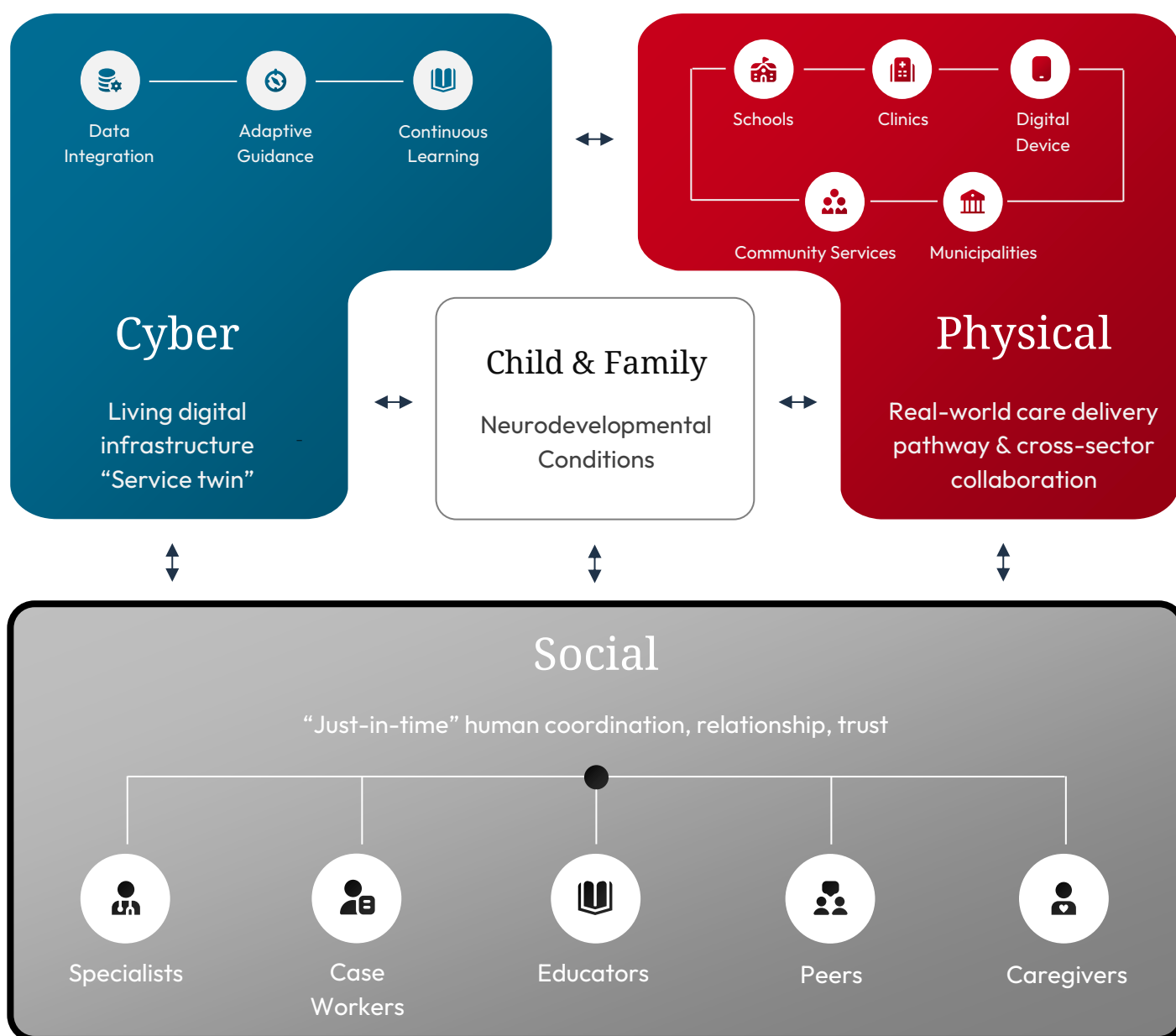
- Society 5.0's investment in the cyber-physical layer and rapidly evolving AI assistive technology, especially in fields with high market returns.
- The missing social layer: trust and relationships, especially in fields with high social returns.
- Policy mandates on reasonable accommodations for all individuals and cross-sector coordination with the data linking architecture being built.

The Opportunity

Neurodevelopmental support is a hard social-routing challenge: poorly defined early concerns, dynamic needs, limited resources, and fragmented pathways.

It offers:

- [1] An actionable bottleneck: a break/delay from signals of concern to appropriate support.
- [2] A proving ground for socially intelligent support infrastructure that can evolve with changing needs.
- [3] A real-world test bed to establish evidence-based services, products, and coordination.



Socially intelligent support infrastructure is a system that **hears** uncertain, pre-diagnostic concerns expressed across platforms people already use, **organizes** those signals into a coherent picture of need, **matches** each family to the tailored combination of supports, and **learns** from measured outcomes.

This requires an integrated build of 1) a **live service map**, 2) a **navigator** that senses need and adapts at each step, 3) a **coordination** layer that connects data and people with consent, and 4) a **governance** with common metrics and specialists-in-the-loop. **Each component creates value but works only when linked to the others.**

The Roadmap & Investment

Phase 0 — 6 months (Completed before funding begins)

Design Sprint

- Aligns partners & governance
- Define shared metrics & ethics guardrails
- Map current vs. target family journeys
- Build MVPs



Phase 1 — Years 1-3

Build & Integrate

- Service twin
- Navigator
- Local ecosystem maps
- Local actors

Pilot (2 regions) & Learn

- Smart city onboarding
- Rural-district introduction

Refine & Judge

- Outcome signals
- Data integration lessons
- Human handoff experience



Phase 2 — Years 4-5 (contingent on Phase 1 success)

Stress & Adapt

- Evaluate longer-term impact
- Expand implementation
- Stress test in a large city

Scale & Sustain

- Assess portability
- Compare outcomes across regions
- Prepare to transition



Phase 1 - Early Outcomes

Budget: US\$12.7M/約 19.1 億円

1 min

Next support step options

1 month

Next support step action

50%↑

Completed handoffs

Phase 2 - Longer-Term Impact

Budget: US\$8.6M/約 12.9 億円

<12 weeks

Support implementation leading to better child-environment fit

3x more

Children supported before crisis, compared to the baseline per region

Total potential program budget: US\$21.3M/約 32.0 億円

MINASA uses a staged, evidence-driven roadmap. The data infrastructure and human networks will be built in parallel. The dual-site design reduces risks by testing the system in varied environments with different focuses. The long-run ambition is 90% of children supported with 10x more support capacity at national scale and cross-domain extension into other socially complex service delivery challenges.

MINASA Ecosystem CPS Layers



CYBER

Adaptive, agentic intelligence about what connections and support work, for whom, and when. Learns, evolves, and improves over time.



PHYSICAL

Coordinated, just-in-time real-world support with shared data and metrics. Leads to greater efficiency and warm handoffs.

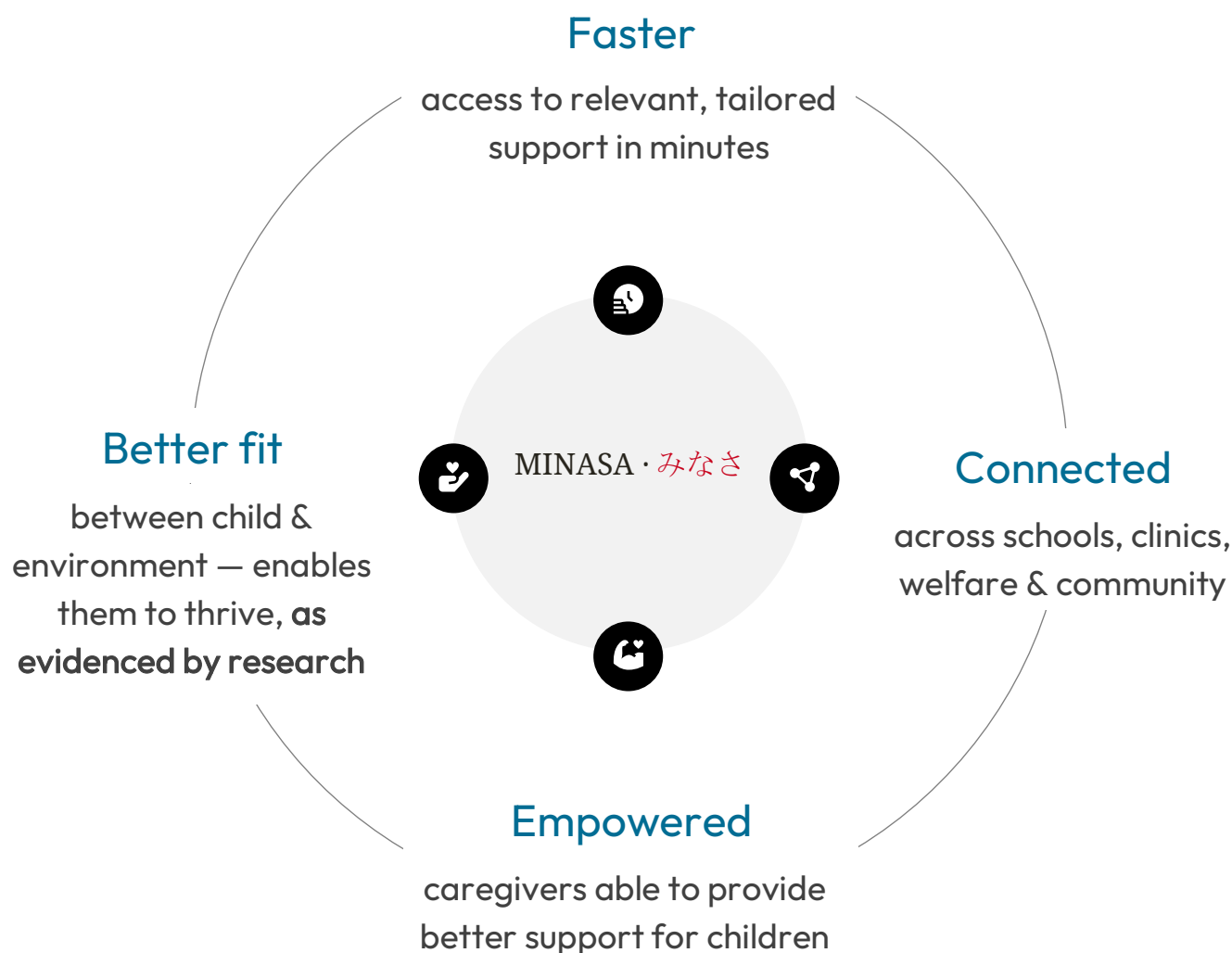


SOCIAL

Varied human support levels, including peer and informal resources. Load-balances care, strengthens relationships, and increases support capacity.

Family Journey with MINASA

Currently, families worry in isolation for years. Once they reach out for support, they carry the navigation burden alone — repeated paperwork, phone calls. **With MINASA, they will experience:**



Core Build Components

The navigator is only the front door. Value comes from linking guidance, local pathways, human handoffs, and learning.

Component	What it builds	Why it matters
Living service twin	Shared map of supports, eligibility, capacity, and pathways.	Makes the local support system searchable, current, and comparable.
Navigation + interfaces	Family and provider guidance, escalation rules, and explanation layer.	Converts need into a clear next-step action while keeping humans in control.
Human coordination layer	Warm handoffs, community connectors, role definitions, and partner onboarding.	Makes support relational, trusted, and completed — not just referred.
Learning + governance	Common metrics, safety rules, privacy architecture, and equity monitoring.	Learns what works while protecting families and improving system planning.

How Key Actors Change

Actor	Before	With MINASA	Result
Child + family	Unclear next steps; repeated storytelling; stress and isolation.	Guided to relevant supports; tries options; shares light feedback.	Faster help, more clarity, better-fit environments, less burden.
School + teacher	Limited visibility; inconsistent accommodations; burden on individual teachers.	Receives practical guidance; coordinates with family; shares observations.	More consistent responses, better classroom fit, improved participation.
Healthcare, welfare + community	Fragmented referrals; informal help invisible; duplicated effort.	Cleaner signals, warmer handoffs, role clarity, better pathway visibility.	Better-matched referrals, earlier help, stronger local networks.
Government / system partner	Limited visibility into needs, gaps, and outcomes.	Maps pathways and gaps; monitors uptake; tests common metrics.	Better planning, allocation, equity monitoring, continuous learning.

Governance, Ethics, and Risk Mitigation

MINASA measures lightly, reviews critical cases with humans, protects anonymity, verifies information, and keeps families safe. **MINASA is not** an app-first program, a diagnostic gatekeeper, a replacement for human judgment, or a surveillance system — it is a consented, closed-loop support pathway coupled to real-world action.

Top Safeguards

Risk area	Guardrail	What to monitor
Over-automation	AI gives options and explanations; humans retain authority; high-risk cases require review.	Review rates, override rates, family-rated relevance and trust.
Privacy and anonymity	Opt-in consent, minimum-necessary data, anonymized learning layer, one-click withdrawal.	Consent, withdrawal, complaints, audits.
Information accuracy	Dual-source verification, provider self-updates, scheduled audits, report-an-error.	Verified records, corrections, failed referrals.
Safety	Safety red lanes, escalation rules, clinical/welfare review, explainability.	Safety incidents, escalation accuracy, subgroup performance.
Bias and inequity	Equity monitoring by region, income, language, age, and access; continuous bias review.	Support completion by subgroup, gap reduction.

Collaboration Model

MINASA builds on a constellation of complementary programs, technical foundations, implementation methods, and policy frameworks.

The goal is not to replace existing services or create another app, but to coordinate them, learn from them, and make them easier for families and local systems to use. This allows MINASA to be sustainable and culturally responsive.

Supplemental Data & References

Scale of unmet need in Japan

1 in 11	children in Japanese regular classrooms (8.8%) show learning or behavioral difficulties consistent with neurodevelopmental conditions — up from 6.5% a decade earlier; most are not formally identified for support, and 43% receive no individual accommodation at school (MEXT 2022 survey).
2.6 years	is the mean lag between a parent's first concern about ADHD symptoms and the child's first clinic visit — lengthened further by young child age, behavioral problems, lower maternal education, and difficulty identifying the right institution (Yamauchi, Fujiwara & Okuyama, Community Mental Health Journal, 2015).
School refusal	is strongly linked to neurodevelopmental conditions, and a large share of affected children remain invisible to the support system until after they have stopped attending school (Inoue, Seishin Igaku, 2024).
Cascade risk	youth with ASD and ADHD traits show substantially elevated suicide-related behavior — yet positive childhood experiences (supportive relationships, safe environments) significantly reduce that risk, especially for those with high ADHD traits (Adachi, Takahashi & Mori, Frontiers in Psychiatry, 2025).

Policy & infrastructure tailwinds

- [1] **Reasonable accommodation** is legally mandated. The amended Act for Eliminating Discrimination against Persons with Disabilities took effect in April 2024, making accommodation obligatory for private businesses as well as government ([Cabinet Office White Paper 2024](#)).
- [2] **Children and Families Agency** was founded in April 2023 as a cross-ministry agency coordinating child-centered policy ([cfa.go.jp](#)).
- [3] **Data sharing** is a national priority. The Digital Agency's Municipal DX dashboard and one-stop service-counter program is building the API-linked architecture ([Digital Agency — Municipal DX](#)).
- [4] **Society 5.0 • Cabinet Office** — Cyber-Physical Systems vision is a national framework integrating cyberspace and physical space ([www8.cao.go.jp/cstp/english/society5_0/index.html](#)).
- [5] **AITeC • AIST** hosts working groups on retrieval-grounded LLM standards, evaluation, and Japanese-language AI governance, and provides technical assistance ([www.ai-tech-c.jp](#)).

Recent grants and programs also emphasize the importance of soft human infrastructure development and maintenance.

Complementary efforts are underway internationally, including coordinating digital tools for ADHD, interoperable data ecosystem design, CPS platform development, community-based approaches, and global efforts to improve child mental health. The program lead collaborates with several of these efforts and has received ARPA-style program development training and support from the Renaissance Philanthropy network.



Bilingual Proposals & Slides
日英提案書・プレゼン



Make every child count. MINASA translates the Japanese tradition of integrated community care into a modern, socially intelligent support infrastructure. Connections enabled by MINASA are the active ingredient to unlock the potential & productivity of the 8.8% of children and their caregivers.

Emi Furukawa, PhD

Senior Staff Scientist

Okinawa Institute of Science and Technology (OIST) Children's Research Center

@ furukawa@oist.jp

Program concept developed as part of Big if True Science (BiTS) Accelerator Program.

Special thanks to the mentors, advisors, program team, and many stakeholders who generously shared their time, expertise, and perspectives throughout MINASA's development.